

Unit - III

Linear Transformation.

Linear Transformation: function (mapping) T from vector space V to vector space (W) .

$(T: V \rightarrow W)$ is said to be a linear transformation if $T(u+v) = T(u) + T(v)$ & $T(cu) = cT(u)$ (preserves addition & scalar multiplication)
for all $u, v \in V$, $c \in \mathbb{R}$

Ex:- Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (x+y, 2z)$.

Let $u, v \in \mathbb{R}^3$

$$\Rightarrow u = (x_1, y_1, z_1), v = (x_2, y_2, z_2)$$

$$u+v = (x_1+x_2, y_1+y_2, z_1+z_2)$$

$$T(u+v) = (x_1+x_2+y_1+y_2, 2(z_1+z_2)) \rightarrow \textcircled{1}$$

$$T(u) = (x_1+y_1, 2z_1)$$

$$T(v) = (x_2+y_2, 2z_2)$$

$$T(u) + T(v) = (x_1+y_1+x_2+y_2, 2(z_1+z_2)) \rightarrow \textcircled{2}$$

from $\textcircled{1}$ & $\textcircled{2}$

$$T(u+v) = T(u) + T(v)$$

Let $c \in \mathbb{R}$ $cu = c(x_1, y_1, z_1) = (cx_1, cy_1, cz_1)$
 $T(cu) = (cx_1 + cy_1, 2cz_1)$

$$\text{Let } c \in \mathbb{R}, \quad cu = c(x_1, y_1, z_1) = (cx_1, cy_1, cz_1)$$

$$\begin{aligned} T(cu) &= T(cx_1, cy_1, cz_1) \\ &= (cx_1 + cy_1, 2cz_1) \rightarrow (3) \end{aligned}$$

$$\begin{aligned} cT(u) &= c(x_1 + y_1, 2z_1) \\ &= (cx_1 + cy_1, 2cz_1) \rightarrow (4) \end{aligned}$$

from (3) & (4)

$$\underline{T(cu) = cT(u)}$$

$\therefore T$ is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

① Note:- A function $T: V \rightarrow W$ is a linear transformation if and only if $T(cu + v) = cT(u) + T(v)$ for all $c \in \mathbb{R}$ & $u, v \in V$

② $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$T(x, y) = xy$$

$\Rightarrow$ Let $u, v \in \mathbb{R}^2$

$\Rightarrow u = (x_1, y_1) \quad v = (x_2, y_2) \quad (\text{Let } c \in \mathbb{R})$

~~$T(cu + v) = cu = c(x_1, y_1)$~~

$$u+v = (x_1 + x_2, y_1 + y_2)$$

$$T(u+v) = (x_1 + x_2)(y_1 + y_2) \rightarrow \textcircled{1}$$

$$T(u) = x_1 y_1$$

$$T(v) = x_2 y_2$$

$$T(u) + T(v) = x_1 y_1 + x_2 y_2 \rightarrow \textcircled{2}$$

from $\textcircled{1}$ & $\textcircled{2}$

$$T(u+v) \neq T(u) + T(v)$$

$\therefore T$ is not a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}$.

NOTE:- If T is a linear transformation from V to W then,

$$T(0) = 0$$

3) A function $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined as
 $T(x, y, z) = (x+1, y+z)$

$$\begin{aligned} T(0, 0, 0) &= (0+1, 0+0) \\ &= (1, 0) \neq (0, 0) \end{aligned}$$

$\therefore T$ is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

① check whether the functions are linear transformations or not.

① $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by
 $T(x_1, x_2, x_3) = (2x_1 - 3x_2, x_1 + 4, 5x_3)$

$$\Rightarrow T(0, 0, 0) = (2(0) - 3(0), 0 + 4, 5(0)) \\ = \underline{(0, 4, 0)} \neq (0, 0, 0)$$

$\therefore T$ is not a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$.

② $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by
 $T(x, y, z) = (x+y, y+z)$ $T(cu+rv) = cT(u) + rT(v)$

$$\Rightarrow \text{Let } u, v \in \mathbb{R}^3 \Rightarrow \begin{cases} u = (x_1, y_1, z_1) \\ v = (x_2, y_2, z_2) \end{cases}$$

$$cu = c(x_1, y_1, z_1) = (cx_1, cy_1, cz_1) \\ cu + v = (cx_1, cy_1, cz_1) + (x_2, y_2, z_2) \\ = (cx_1 + x_2, cy_1 + y_2, cz_1 + z_2)$$

$$T(cu+v) = (cx_1 + x_2 + cy_1 + y_2, cy_1 + y_2 + cz_1 + z_2)$$

$$cT(u) = c(x_1 + y_1, y_1 + z_1) = (cx_1 + cy_1, cy_1 + cz_1)$$

$$cT(u) + T(v) = (cx_1, cy_1, cz_1) + (x_2, y_2, z_2) \\ = (cx_1 + x_2, cy_1 + y_2, cz_1 + z_2) \quad \text{--- (2)}$$

from (1) & (2)

$$\underline{T(cu + v) = cT(u) + T(v)}$$

$$cT(u) + T(v) = (cx_1 + cy_1, cy_1 + cz_1) (x_2 + y_2, y_2 + z_2) \\ = (cx_1 + cy_1 + x_2 + y_2, cy_1 + cz_1 + y_2 + z_2) \\ \Downarrow \\ \text{--- (2)}$$

from (1) & (2)

$$\underline{T(cu + v) = cT(u) + T(v)}$$

$\therefore T$ is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$.

③ $T: \overset{\mathbb{R}^3}{V_3(\mathbb{R})} \rightarrow \overset{\mathbb{R}^2}{V_2(\mathbb{R})}$ defined as

$$T(x_1, x_2, x_3) = (x_1 - x_2, x_1 + x_3).$$

$\Rightarrow$ Let $u, v \in \mathbb{R}^3 \Rightarrow u = (x_1, x_2, x_3)$
 $v = (y_1, y_2, y_3)$

$$cu = c(x_1, x_2, x_3) = (cx_1, cx_2, cx_3)$$

$$cu + v = (cx_1, cx_2, cx_3) + (y_1, y_2, y_3)$$

$$= (cx_1 + y_1, cx_2 + y_2, cx_3 + y_3)$$

$$T(cu + v) = (cx_1 + y_1 - cx_2 - y_2, cx_1 + y_1 + cx_3 + y_3) \quad \text{--- (1)}$$

$$cT(u) = (cx_1 - x_2, x_1 + x_3)$$

$$= (cx_1 - cx_2, cx_1 + cx_3)$$

$$cT(u) + T(v) = (cx_1 - cx_2, cx_1 + cx_3)(y_1 - y_2, y_1 + y_3)$$

$$= (cx_1 - cx_2 + y_1 - y_2, cx_1 + cx_3 + y_1 + y_3)$$

from ① & ②

$$T(u+v) = cT(u) + T(v)$$

$\therefore T$ is a linear transformation from $V_3(\mathbb{R}) \rightarrow V_2(\mathbb{R})$.

④ $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T(x, y) = (x+y, x)$

$\Rightarrow$ let $u, v \in \mathbb{R}^2 \Rightarrow u = (x_1, y_1)$
 $v = (x_2, y_2)$

$T(cu+v) \& cu = (cx_1, cy_1)$

$cu+v = (cx_1, cy_1) + (x_2, y_2)$

$= (cx_1 + x_2, cy_1 + y_2)$

$T(cu+v) = (cx_1 + x_2 + cy_1 + y_2, cx_1 + x_2)$
①

$cT(u) = c(x_1 + y_1, x_1)$

$= (cx_1 + cy_1, cx_1)$

$cT(u) + T(v) = (cx_1 + cy_1, cx_1) + (x_2 + y_2, x_2)$

$= (cx_1 + cy_1 + x_2 + y_2, cx_1 + x_2)$
②

from ① & ② $T(au+av) = aT(u) + T(v)$
 $\therefore T$ is a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}$.

Range space & null space of linear transformation.

Let $T: V \rightarrow W$ will be a linear transformation then the range of function T is defined by $\text{range}(T)$. It is defined as image space of T .

$$\text{i.e., } \text{range}(T) = \{T(u) \mid u \in V\}$$

The null space or kernel of T is the set of vectors in V which are mapping to zero vector in W , it is denoted by $\text{null}(T)$.

$$\text{i.e., } \text{null}(T) = \{u \in V \mid T(u) = 0\}$$

NOTE:- If $T: V \rightarrow W$ is a linear transformation then $\text{null}(T)$ is a subspace of V and $\text{range}(T)$ is a subspace of W .

ex:- $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(x, y) = (x+y, 0)$ be a ~~subspace of V~~ linear transformation.

$$\Rightarrow \text{Range}(T) = \{T(u) \mid u \in \mathbb{R}^2\}$$

$$= \{ T(x, y) \mid (x, y) \in \mathbb{R}^2 \}$$

$$\text{Range}(T) = \{ (x+y, 0) \mid x, y \in \mathbb{R} \}$$

$$\{ (x+y) (1, 0) \mid x, y \in \mathbb{R} \}$$

$$= \{ k (1, 0) \mid k \in \mathbb{R} \}$$

$$\text{Range}(T) = \text{span} \{ (1, 0) \}$$

$$\text{Basis of range}(T) = \{ (1, 0) \}$$

$$\dim(\text{range}(T)) = 1$$

$$\text{Nul space}(T) = \{ u \in \mathbb{R}^2 \mid T(u) = 0 \}$$

$$\{ (x, y) \in \mathbb{R}^2 \mid T(x, y) = (0, 0) \}$$

$$\{ (x, y) \in \mathbb{R}^2 \mid (x+y, 0) = (0, 0) \}$$

$$x+y=0$$

$$x=-y \text{ or }$$

$$y=-x$$

$$\{ (x, y) \mid y = -x, x, y \in \mathbb{R} \}$$

$$\{ (x, -x) \mid x \in \mathbb{R} \}$$

$$T(2, -2) = 2-2$$

$$= 0$$

$$\{x(1, -1) \mid x \in \mathbb{R}\}$$

$$\text{span}\{(1, -1)\}$$

Basis of $\text{nul}(T)$ is $\{(1, -1)\}$

$$\underline{\underline{\dim(\text{nul}(T)) = 1}}$$

① Let a linear Transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined from $T(x, y, z) = (y-x, y-z)$ find the basis & the dimension of nullspace & range space (T) .

⇒ matrix linear transformation T with respect to std. basis is,

Std. basis of $\mathbb{R}^3$ is $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

$$T(1, 0, 0) = (-1, 0)$$

$$T(0, 1, 0) = (1, 1)$$

$$T(0, 0, 1) = (0, -1)$$

$$A = \begin{bmatrix} x & y & z \\ -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

$$\text{Range}(T) = \text{col}(A)$$

$$\text{Nul}(T) = \text{nul}(A)$$

$$\text{Range}(T) = \text{col}(A)$$

$$= \text{span}\{(-1, 0), (1, 1), (0, -1)\}$$

Basis of range space of $T =$ Basis of col
Pivot col^n

$$= \{(-1, 0), (1, 1)\}$$

$$\therefore \dim(\text{Range}(T)) = 2$$

$\text{null}(T) = \text{null}(A) = \text{sol}^n$ of set of $Ax = 0$
To find x , solve $Ax = 0$

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-x + y = 0$$

$$y - z = 0$$

$$y - x = 0 \Rightarrow \underline{y = x}$$

$$\underline{y = z}$$

$$\therefore \underline{x = y = z = k} \text{ (free variable)}$$

$$x = (x, y, z) = (k, k, k)$$

$$\text{Nul}(T) = \{(k, k, k) \mid k \in \mathbb{R}\}$$

$$\{k(1, 1, 1) \mid k \in \mathbb{R}\}$$

$$\therefore \text{span}\{(1, 1, 1)\}$$

Rank - nullity theorem.

Let $T: V \rightarrow W$ be a linear transformation, then $\dim(V) = \text{Rank}(T) + \text{nullity}(T)$

Q] Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a L.T defined by $T(x, y, z) = (x+y+z, x+2y+3z)$ then find rank - nullity of T .

$\Rightarrow$ Rank of the L.T, matrix of T w.r.t std. basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ & $\{(1, 0), (0, 1)\}$ is $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$

$$\text{Range}(T) = \text{col}(A) = \text{span}\{(1, 1), (1, 2), (1, 3)\}$$

$$R_2 \rightarrow R_2 - R_1$$

Ans

$$A \sim \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

Pivot cols of A are $(1,1)$ $(1,2)$

$\therefore$ Basis of $\text{col}(A) = \{(1,1) (1,2)\} = \text{Basis of range}(T)$.

$$\therefore \text{Rank}(T) = \dim(\text{range}(T)) = 2$$

from rank + nullity theorem,

$$\dim(\mathbb{R}^3) = \text{Rank}(T) + \text{nullity}(T)$$

$$3 = 2 + \text{nullity}(T)$$

Q] Verify Rank nullity theorem for the linear transformation defined by $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ - codomain
 $T(x, y, z) = (x, 0, x - y + z)$ domain

⇒ matrix of $L.T$ w.r.t standard basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 1 \end{bmatrix}$$

Range $(T) = \text{col}(A) = \text{span columns of } A$ $\neq$ dependent
 $= \text{span} \{(1, 0, 1), (0, 0, -1), \cancel{(0, 0, 1)}\}$

Basis of $(\text{Range}(T)) = \text{Basis of } \text{col}(A)$
 $= \{(1, 0, 1), (0, 0, -1)\}$

$\text{Rank}(T) = \dim(\text{range}(T)) = \underline{2}$

$\text{Nul}(T) = \{(x, y, z) \mid T(x, y, z) = 0 = (0, 0, 0)\}$

$\{(x, y, z) \mid (x, 0, x - y + z) = (0, 0, 0)\}$

$\{x, y, z \mid x = 0, 0 = 0, x - y + z = 0\}$

$$\{(x, y, z) \mid x=0, 0-y+z, y=z\}$$

$$(\text{Null}(T)) = \{(x, y, z)\}$$

$$\text{Null}(T) = \{(0, y, y) \mid y \in \mathbb{R}\}$$

$$= \{y(0, 1, 1) \mid y \in \mathbb{R}\}$$

$$= \text{span} \{(0, 1, 1)\}$$

Basis of

$$\text{Null}(T) = \{(0, 1, 1)\}$$

$$\text{Nullity}(T) = \dim(\text{Null}(T)) = 1$$

$$\therefore \text{Rank}(T) + \text{Nullity}(T)$$

$$2 + 1 = \underline{3} = \dim(\mathbb{R}^3)$$

(2) find range space, null space Rank & nullity of L.T of T , $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x+y, x-y, x+2z)$.

$\Rightarrow$ matrix of L.T T w.r.t standard basis $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

x y z independent

$$\text{Range}(T) = \text{col}(A)$$

$$= \text{span of col}(A)$$

$$= \text{span} \{ (1, 1, 0), (1, -1, 0), (0, 0, 2) \}$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_2 \rightarrow R_2 - R_1$$

$$A \sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$$

$$R_3 \rightarrow 2R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 1 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\text{Basis of Range}(T) = \text{Basis of col}(A)$$

$$= \{ (1, 1, 0), (1, -1, 0), (0, 0, 2) \}$$

$$\text{Rank}(T) = \dim(\text{Range}(T)) = \underline{3}$$

$$\text{Nul}(T) = \{ (x, y, z) \mid T(x, y, z) = 0 = (0, 0, 0) \}$$

$$\{ (x, y, z) \mid (x+y, x-y, x+2z) = (0, 0, 0) \}$$

$$\{ (x, y, z) \mid \begin{matrix} x+y=0, & x-y=0, & x+2z=0 \end{matrix} \}$$

$$\begin{matrix} x = -y \\ x + x = 0 \\ y = 0 \end{matrix}$$

$$2x = 0 \\ x = 0$$

$$z = 0$$

$$\text{Nul}(T) = \{ (0, 0, 0) \}$$

$$\text{Basis of Nul}(T) = \{ \cancel{(0, 0, 0)} \}$$

$$\text{Nullity}(T) = \dim(\text{Nul}(T)) = 0$$

$$\begin{aligned} &= \text{Rank}(T) + \text{Nullity}(T) \\ &= 3 + 0 = 3 = \dim(\mathbb{R}^3) \end{aligned}$$

Matrix of the linear transformation relative to the basis:

Let $T: V \rightarrow W$ be a linear transformation
Suppose

$B = \{u_1, u_2, \dots, u_n\}$ is a basis of V

$B' = \{v_1, v_2, \dots, v_m\}$ is a basis of W

Then the vectors $T(u_1), T(u_2), \dots, T(u_n)$ in W , can be written as a linear (transformation) combination on the basis that is...

Defn:- The transpose of the matrix of coefficients in the above eq's, denoted by $[T]_{B, B'}$ is called the matrix representation of T relative to the bases B and B' .

$$\text{i.e., } [T]_{B, B'} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

using the coordinate vector notation, the matrix representation of T relative to the bases B & B' may be written in the form,

$$[T]_{B, B'} = [[T(u_1)]_{B'}, [T(u_2)]_{B'}, \dots, [T(u_n)]_{B'}]$$

To find matrix of $L.T.$

① find images of $u_1, u_2, \dots, u_n$
 $T(u_1), T(u_2), \dots, T(u_n)$

② coordinate vectors of $T(u_1), T(u_2), \dots, T(u_n)$
 relative to basis $B' = \{v_1, v_2, \dots, v_n\}$

① find the matrix transformation of $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined $T(x, y) = (x, -y)$.

$$B_1 = \{(1, 1), (1, 0)\} \text{ } B_2 = \{(2, 3), (4, 5)\}$$

$$\Rightarrow \text{let } u_1 = (1, 1), u_2 = (1, 0) \rightarrow \text{domain}$$

$$v_1 = (2, 3), v_2 = (4, 5) \rightarrow \text{codomain}$$

$$T(u_1) = T(1, 1) = (1, -1) //$$

$$T(u_1) = a_{11}v_1 + a_{21}v_2$$

$$T(u_1) = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = a_{11} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + a_{21} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$2a_{11} + 4a_{21} = 1$$

$$3a_{11} + 5a_{21} = -1$$

$$a_{11} = -4.5$$

$$a_{21} = \frac{2.5}{2} = 5/2$$

$$T(u_2) = T(1, 0) = (1, 0) //$$

$$= a_{12}v_1 + a_{22}v_2$$

$$T(u_2) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = a_{12} \begin{bmatrix} 2 \\ 3 \end{bmatrix} + a_{22} \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$2a_{12} + 4a_{22} = 1$$

$$3a_{12} + 15a_{22} = 0$$

$$S = 10$$

$$a_{12} = -2.5 //$$

$$a_{22} = 1.5 //$$

$$1 \times 0 + 10$$

$$1 \times 0 + 2$$

matrix of L.T is, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$[T]_{B_1, B_2} = \begin{bmatrix} -4.5 & -2.5 \\ 2.5 & 1.5 \end{bmatrix}$$

② find the matrix transformation $T: V_2 \rightarrow V_3$ defined by $T(x, y) = (x+y, x, 3x-y)$

$$B_1 = \{(1, 1), (3, 1)\}$$

$$B_2 = \{(1, 1, 1), (1, 1, 0), (0, 0, 0)\}$$

$$\Rightarrow u_1 = (1, 1) \quad u_2 = (3, 1)$$

$$v_1 = (1, 1, 1) \quad v_2 = (1, 1, 0) \quad v_3 = (0, 0, 0)$$

$$S = 10$$

$$T(u_1) = T(1, 1) = (2, 1, 2)$$

$$= a_1 v_1 + a_2 v_2 + a_3 v_3$$

$$\begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} = a_1 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + a_3 \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$a_1 + a_2 + a_3 = 2 \quad \text{--- (1)}$$

$$a_1 + a_2 = 1 \quad \text{--- (2)}$$

$$\boxed{a_1 = 2}$$

$$a_1 + a_2 = 1$$

$$2 + a_2 = 1$$

$$\boxed{a_2 = -1}$$

$$2a_3 + 2(-1) = 2$$

$$\boxed{a_3 = 1}$$

$$T(u_2) = T(3, 1) = (4, 3, 8)$$

$$(u_2)_T = b_1 v_1 + b_2 v_2 + b_3 v_3$$

$$\begin{bmatrix} 4 \\ 3 \\ 8 \end{bmatrix} = b_1 \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} + b_2 \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} + b_3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$(0, 0, 1) = b_1 + b_2 + b_3 = 4$$

$$b_1 + b_2 = 3$$

$$\boxed{b_1 = 8}$$

$$8 + b_2 + b_3 = 4$$

$$\begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} 8 + \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix} b_2 + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} b_3 = \begin{bmatrix} 4 \\ 3 \\ 8 \end{bmatrix}$$

$$\boxed{b_2 = -5}$$

$$8 - 5 + b_3 = 4$$

$$b_3 = 4 - 3$$

$$\boxed{b_3 = 1}$$

$$[T]_{B_1, B_2} = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{bmatrix} = \begin{bmatrix} 2 & 8 \\ -1 & -5 \\ 1 & 1 \end{bmatrix}$$

④ find the M.T from $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ $T(x, y, z) = (x - y + 2z, 3x + y)$

$$B_1 = \{(1, 1, 1), (1, 2, 3), (1, 0, 0)\}$$

$$B_2 = \{(1, 1), (1, -1)\}$$

Dot product / scalar product:

Let $u, v \in \mathbb{R}^n$, then the dot product of u & v is denoted by $u \cdot v$, defined as,
 $u \cdot v = u^T v$.

$$\text{If } u = (u_1, \dots, u_n) \quad v = (v_1, v_2, \dots, v_n)$$

$$\text{then } u \cdot v = u^T v = [u_1 \ u_2 \ \dots \ u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$= u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

$$= \underline{v \cdot u}$$

Ex:- Let $u = (1, -2, 3, 5)$ & $v = (1, -1, 0, 5)$

$$u \cdot v = (1 \times 1) + (-2 \times -1) + (3 \times 0) + (5 \times 5)$$

$$= 1 + 2 + 25$$

$$= 28 = v \cdot u$$

$$\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + v_2^2 + v_3^2 + v_4^2}$$

$$= \sqrt{1 + 1 + 0 + 25}$$

$$= \sqrt{27} = 3\sqrt{3}$$

orthogonal vectors: Two vectors u and v are said to be orthogonal or perpendicular vector if dot product of u & v i.e. $u \cdot v = 0$ or the angle between two vectors should be equal to 90° .

orthogonal set.

A set $\{u_1, u_2, \dots, u_n\}$ of vectors is said to be orthogonal if $u_i \cdot u_j = 0$ for all i, j with $i \neq j$.

Note:- Let $\{u_1, u_2, \dots, u_n\}$ be an orthogonal set of non-zero vectors then,

$\{u_1, u_2, u_3, \dots, u_n\}$ is linearly independent.

eg:- Let $u_1 = (3, 1, 1)$ $u_2 = (-1, 2, 1)$ $u_3 = (-1/2, -2, 7/2)$

$$u_1 \cdot u_2 = (3 \times -1) + (1 \times 2) + (1 \times 1) = -3 + 2 + 1 = 0$$

$$u_1 \cdot u_3 = (3 \times -1/2) + (1 \times -2) + (1 \times 7/2) = -3/2 - 2 + 7/2 = 0 //$$

$$u_2 \cdot u_3 = 1/2 - 4 + 7/2 = 0 //$$

Orthonormal set: An orthogonal set $\{v_1, v_2, \dots, v_n\}$ of vectors is orthonormal if $\|v_i\| = 1$, for all i .

Ex:- Let $u_1 = (3, 1, 1)$, $u_2 = (-1, 2, 1)$, $u_3 = (-1/2, -2, 7/2)$

$$\|u_1\| = \sqrt{3^2 + 1^2 + 1^2} = \sqrt{9 + 1 + 1} = \underline{\underline{\sqrt{11}}}$$

$$\frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{11}} (3, 1, 1) = v_1$$

unit vector

$$\|u_2\| = \sqrt{1^2 + 2^2 + 1^2} = \underline{\underline{\sqrt{6}}}$$

$$\frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{6}} (-1, 2, 1) = v_2$$

$$\|u_3\| = \sqrt{(-1/2)^2 + (-2)^2 + (7/2)^2}$$

$$\sqrt{0.25 + 4 + 12.25}$$

$$\sqrt{16.5} = \underline{4.06}$$

$$\frac{u_3}{\|u_3\|} = \frac{1}{4.06} (-1/2, -2, 7/2) = v_3$$

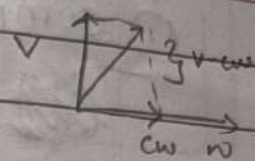
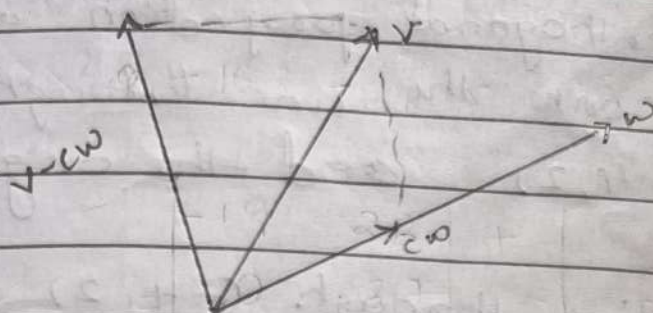
$\{v_1, v_2, v_3\}$ is an orthonormal set

orthogonal projection of vector.

Let w be a vector in $\mathbb{R}^n$. Suppose v is another vector. We seek the "projection of v along w ", which, as indicated in figure, will be the multiple cw of w such that $v - cw$ is orthogonal to w . This means:

$$(v - cw) \cdot w = 0 \quad (\text{or})$$

$$c = \frac{v \cdot w}{w \cdot w}$$



Then the vector $cw = \hat{v} = \text{proj}_w v = \frac{v \cdot w}{w \cdot w} w$ is

called orthogonal projection of v onto w , & the vector $v - \hat{v}$ is called the component of v orthogonal to w .

If k is any non-zero scalar if w is replaced by kw in the defⁿ of $\hat{v}$, then the orthogonal projection of v onto kw is exactly the same as the orthogonal projection of v onto w . Hence this projection is determined by the subspace w spanned by w (that is the line through w & 0). Sometimes $\hat{v}$ is denoted by $\text{proj}_w v$ is called the orthogonal projection of v onto w .

$$\text{i.e., } \hat{v} = \text{proj}_w v = \frac{v \cdot w}{w \cdot w} w$$

① compute orthogonal projection of a vector $(1, 7)$ onto the line through the vector $(-4, 2)$ and the origin.

$\Rightarrow$ let $w =$ line through $(-4, 2) \in w, 0$
 $= \text{span} \{(-4, 2)\}$
 let $v = (1, 7)$

$$\text{proj}_w v = \frac{v \cdot w}{w \cdot w} w$$

$$v \cdot w = (1, 7) \cdot (-4, 2)$$

$$= -4 + 14 = 10$$

$$(w \cdot w = (1, 7) \cdot (1, 7))$$

$$= 1 + 49 = 50$$

$$w \cdot w = +16 + 4$$

$$= 20$$

$$= \frac{10}{20} (-4, 2)$$

$$= \frac{1}{2} (-4, 2)$$

$$= \underline{\underline{(-2, 1)}}$$

- ② let $y = (7, 6)$ and $u = (4, 2)$ find the orthogonal projection of y on u and also write y as sum of two vectors: one is $\text{span}\{u\}$ and one is orthogonal to u .

$$\Rightarrow y = (7, 6) \quad u = (4, 2)$$

$$\text{proj}_u y = \frac{y \cdot u}{u \cdot u} u$$

$$y \cdot u = 28 + 12 = 40$$

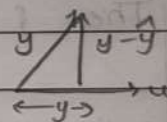
$$u \cdot u = 16 + 4 = 20$$

$$= \frac{40}{20} (4, 2)$$

$$= 2(4, 2)$$

$$\hat{y} = (8, 4) \in \text{span}\{u\}$$

$$y = \hat{y} + (y - \hat{y}) \rightarrow$$



$$y - \hat{y} = (7, 6) - (8, 4)$$

$$= (-1, 2) = \text{orthogonal to } u$$

$$y = \hat{y} + (y - \hat{y})$$

$$= (8, 4) + (-1, 2)$$

$$y = \underline{\underline{(7, 6)}}$$

- ③ compute orthogonal projection of $y = (2, 3)$ on the line through the point $u = (4, -7)$ then write y as the sum of two orthogonal vectors: one as $\text{span}\{u\}$ and one orthogonal to u .

Grahm - Schmidt process:

The gram schmidt process is a simple algorithm for pocketing an orthogonal / orthonormal basis for any non-zero subspace of $\mathbb{R}^n$.

Given a basis $\{x_1, x_2, \dots, x_p\}$ for a non-zero subspace W of $\mathbb{R}^n$ define,

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$v_p = x_p - \frac{x_p \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_p \cdot v_2}{v_2 \cdot v_2} v_2 - \dots - \frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} v_{p-1}$$

Then $\{v_1, v_2, \dots, v_p\}$ is orthogonal vector

Orthonormal Basis:

Orthonormal basis is constructed easily from an orthogonal basis $\{v_1, v_2, \dots, v_p\}$ simply normalize all the v_k .

Q] Let $W = \text{span}\{x_1, x_2\}$ where $x_1 = (3, 6, 0)$, $x_2 = (1, 2, 2)$. Find an orthogonal basis of W by using Gram Schmidt process.

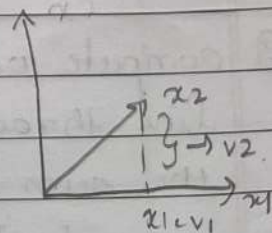
$\Rightarrow$

Let $v_1 = x_1$

$$v_2 = x_2 - \text{proj}_{v_1} x_2$$

$$= x_2 - \frac{v_1 \cdot x_2}{v_1 \cdot v_1} v_1$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \frac{\sqrt{5}}{4\sqrt{3}} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}$$



$$x_2 \cdot v_1 = x_2 \cdot x_1$$

$$= 3 + 12$$

$$= 15$$

$$= \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$\therefore \{v_1, v_2\}$ is an orthogonal basis of w

2) Let $w = \text{span}\{x_1, x_2, x_3\}$ where $x_1 = (1, 1, 1, 1)$, $x_2 = (0, 1, 1, 1)$ & $x_3 = (0, 0, 1, 1)$. find orthonormal basis of w by using Gram Schmidt process.

$\Rightarrow$ Let $x_1 = v_1$

$$v_2 = x_2 - \text{proj}_{v_1} x_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1$$

$0+1+1+1=3$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{3}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -3/4 \\ 1/4 \\ 1/4 \\ 1/4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$v_3 = x_3 - \text{proj}_{v_1} x_3 - \text{proj}_{v_2} x_3$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{2}{3} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$\frac{1}{4} \times \frac{1}{4} (12)$

$$z \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{6} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 - \frac{1}{2} + \frac{1}{6} \\ 0 - \frac{1}{2} - \frac{1}{6} \\ 1 - \frac{1}{2} - \frac{1}{6} \\ 1 - \frac{1}{2} - \frac{1}{6} \end{bmatrix} = \begin{bmatrix} 0 \\ -4/6 \\ 1/3 \\ 1/3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2/3 \\ 1/3 \\ 1/3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore \{v_1, v_2, v_3\}$ is an orthogonal basis of w .

$$\text{let } u_1 = \frac{v_1}{\|v_1\|} = \frac{(1, 1, 1, 1)}{\sqrt{1^2 + 1^2 + 1^2 + 1^2}} = \frac{1}{\sqrt{4}} (1, 1, 1, 1) \\ = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$u_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{(1/4)^2 [(-3)^2 + 1^2 + 1^2 + 1^2]}} \times \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix} \\ = \frac{1}{\frac{1}{4}(\sqrt{12})} \times \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\frac{1}{4} \times 2\sqrt{3}} \times \frac{1}{4} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{2\sqrt{3}} \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \frac{v_3}{\|v_3\|}$$

$$= \frac{1}{\sqrt{\left(\frac{1}{3}\right)^2 + 0 + 4 + 1 + 1}} \times \frac{1}{3} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$\frac{1}{3\sqrt{6}} \times \frac{1}{3} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$$u_3 = \frac{1}{\sqrt{6}} \begin{bmatrix} 0 \\ -2 \\ 1 \\ 1 \end{bmatrix}$$

$\therefore \{u_1, u_2, u_3\}$ is an orthonormal basis

3] Let $W = \text{span}\{x_1, x_2, x_3\}$ where $x_1 = (1, 1, 1, 1)$, $x_2 = (1, -1, 2, 3)$, $x_3 = (1, 2, -3, -4)$. Construct an orthogonal basis of W .

$\Rightarrow$

$$v_1 = x_1$$

$$v_2 = x_2 - \frac{x_2 \cdot v_1 \cdot v_1}{v_1 \cdot v_1}$$

$$x_2 \cdot x_1 = 1 + (-1) + 2 + 3 = 5$$

$$x_1 \cdot x_1 = 4$$

$$= \begin{bmatrix} 1 \\ -1 \\ 2 \\ 3 \end{bmatrix} - \frac{5}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} -1/4 \\ -5/4 \\ 3/4 \\ 7/4 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -1 \\ -5 \\ 3 \\ 7 \end{bmatrix}$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} v_2$$

$$= x_3 - \frac{(-4)}{4} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \frac{1/4(-56)}{1/16(1+81+9+49)} \begin{bmatrix} -1 \\ -9 \\ 3 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ -3 \\ -4 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} + \frac{2}{5} \begin{bmatrix} -1 \\ -9 \\ 3 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 1 + 1 - 2/5 \\ 2 + 1 - 18/5 \\ -3 + 1 + 6/5 \\ -4 + 1 + 14/5 \end{bmatrix} = \begin{bmatrix} 9/5 \\ -3/5 \\ -1/5 \\ -1/5 \end{bmatrix}$$

$\{v_1, v_2, v_3\}$ is an orthogonal basis of W .

4) If $W = \text{span}\{x_1, x_2\}$, where $x_1 = (2, -8, 1)$
 $x_2 = (4, -1, 2)$. construct orthonormal basis for W .

$$\begin{bmatrix} 2 & 4 \\ -8 & -1 \\ 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$$